Notes on lecture 7

Jiten Chandra Kalita and Shyamashree Upadhyay

1 Differentials

1.1 For functions of one variable

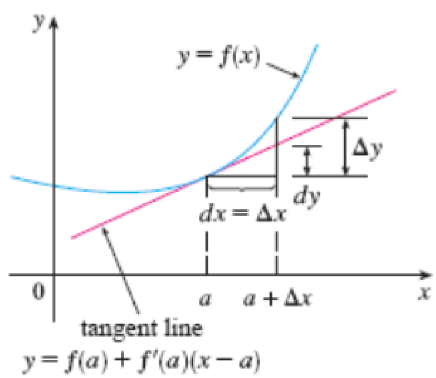

Figure 1.1.1: Geometrical interpretation of the differential (one variable case)

Let $y = f(x)$. Then we know that the **differential** dy of y is given by

$$dy = f'(x)dx.$$

Suppose we are considering the graph of the function $y = f(x)$ and we are also considering the tangent line to this graph at the point $x = a$ (see Figure 1.1.1). Let us also consider a point $a + \Delta x$ on the x-axis, in a neighbourhood of the point a. Then the increment of the independent variable x is equal to Δx. We will write Δx as dx. Also, the increment of the dependent variable y is given by

$$\Delta y = f(a + \Delta x) - f(a).$$

We also know that the equation of the tangent line is given by

$$y - f(a) = f'(a)(x - a).$$

That is,

$$y = f(a) + f'(a)(x - a).$$

Now, if we consider $a + \Delta x$ as x, then the above equation becomes

$$y = f(a) + f'(a)\Delta x.$$

Let us define another function $L(x)$ of the independent variable x as follows:

$$L(x) := f(a) + f'(a)\Delta x.$$

Then we have,

$$L(x) - f(a) = f'(a)\Delta x = f'(a)dx.$$

If we take the variable point x as $a+\Delta x$, then the **differential** dy is nothing but the distance between $L(x)$ and $f(a)$. That is,

$$dy = L(x) - f(a) = f'(a)dx.$$

Whereas, Δy (the increment in the dependent variable y) is equal to $f(x) - f(a) = f(a+\Delta x) - f(a)$ (as $x = a+\Delta x$). In other words, the differential dy is the distance between $f(a)$ and the point on the tangent line corresponding to $a+\Delta x$. And Δy is the distance between $f(a)$ and the point on the graph of the function corresponding to $a+\Delta x$. This is the geometrical interpretation of the differential in the case of functions of one variable. Please see Figure 1.1.1.

1.2 For functions of two variables

Let $f(x, y)$ be a function of two independent variables x and y. Let Δx and Δy be the increments in the independent variables x and y respectively. We will write Δx and Δy as dx and dy respectively.
We know that the equation of the tangent plane to the surface $z = f(x, y)$ at the point (a, b, c) (where $c = f(a, b)$) is given by

$$z - c = f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

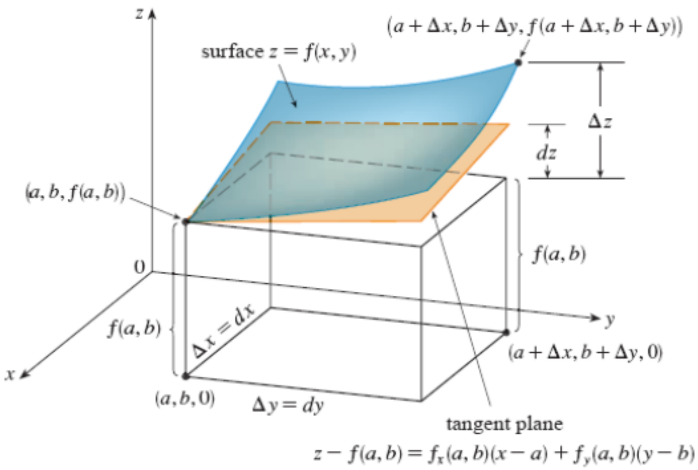

Figure 1.2.1: Geometrical interpretation of the differential (two variables)

If we take the variable points x and y as $a+\Delta x$ and $b+\Delta y$ respectively, then the above equation of the tangent plane becomes

$$z - c = f_x(a,b)\Delta x + f_y(a,b)\Delta y.$$

In other words, the equation of the tangent plane becomes

$$z - f(a,b) = f_x(a,b)dx + f_y(a,b)dy.$$

This difference $z - f(a,b)$ (which is equal to $f_x(a,b)dx + f_y(a,b)dy$) is called the **differential** or the **total differential** of z, and is denoted by dz.
For a variable point (x,y), we will have

$$dz = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy.$$

As in the case of one variable, here also we have:

$$\Delta z = f(a+\Delta x, b+\Delta y) - f(a,b),$$

which is the distance between $f(a,b)$ and the point on the surface corresponding to $(a+\Delta x, b+\Delta y)$. Whereas, dz is the distance between $f(a,b)$

and the point on the tangent plane corresponding to $(a+\Delta x, b+\Delta y)$. This is the geometrical interpretation of the differential in the case of functions of 2 variables. Please see Figure 1.2.1.
Note: Using the definition of the total differential dz, the differentiability of $f(x,y)$ may be redefined as follows:
A function $f(x,y)$ of two variables x and y is said to be differentiable at (a,b) if

$$\lim_{(\Delta x,\Delta y)\to(0,0)} \frac{\Delta z - dz}{\sqrt{(\Delta x)^2+(\Delta y)^2}} = 0.$$

2 The chain rule

Let us first consider a differentiable function of one variable, say $y=f(x)$. Suppose that x again is a differentiable function of another variable t. Then we can write y as a function of t and we can find the ordinary derivative of y with respect to t, and it is given by:

$$\frac{dy}{dt} = \frac{dy}{dx}\frac{dx}{dt}.$$

Now, let z be a differentiable function of 2 variables. Say, $z=f(x,y)$. Also assume that each of x and y are differentiable functions of another variable t. That is,

$$z = f(x,y) = f(x(t), y(t)).$$

Now, if f is a differentiable function, then it follows from the definition of differentiability that

$$\Delta z = \frac{\partial f}{\partial x}\Delta x + \frac{\partial f}{\partial y}\Delta y + \epsilon_1 \Delta x + \epsilon_2 \Delta y, \tag{1}$$

where $\epsilon_1, \epsilon_2 \to 0$ as $(\Delta x, \Delta y) \to (0,0)$. Now, as x is a function of t, we have

$$\Delta x = x(t+\Delta t) - x(t).$$

Therefore,

$$\lim_{\Delta t\to 0}\frac{\Delta x}{\Delta t} = \lim_{\Delta t\to 0}\frac{x(t+\Delta t)-x(t)}{\Delta t} = \frac{dx}{dt}.$$

Similarly,

$$\lim_{\Delta t\to 0}\frac{\Delta y}{\Delta t} = \frac{dy}{dt}.$$

Again, from equation (1), we can write

$$\frac{\Delta z}{\Delta t} = \frac{\partial f}{\partial x}\frac{\Delta x}{\Delta t} + \frac{\partial f}{\partial y}\frac{\Delta y}{\Delta t} + \epsilon_1 \frac{\Delta x}{\Delta t} + \epsilon_2 \frac{\Delta y}{\Delta t}. \tag{2}$$

Observe that since x is a continuous function of t, therefore

$$\Delta x = x(t + \Delta t) - x(t) \to 0 \text{ as } \Delta t \to 0.$$

Similarly for Δy. This implies that as $\Delta t \to 0$, we have $(\Delta x, \Delta y) \to (0, 0)$ and hence $\epsilon_1, \epsilon_2 \to 0$. So by taking limits in equation (2) as $\Delta t \to 0$, we get:

$$\frac{dz}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}.$$

This is the chain rule for a function of 2 variables, where each of those 2 variables is again a function of a single variable t.

Example 2.0.1. Let $z = x^2 y + 3xy^4$, where $x = \sin 2t$ and $y = \cos t$. Find $\frac{dz}{dt}_{|t=0}$.

Using the chain rule, we have:

$$\frac{dz}{dt} = (2xy + 3y^4)(2\cos 2t) + (x^2 + 12xy^3)(-\sin t).$$

So at $t = 0$, we have:

$$x = 0, y = 1,$$

and hence

$$\frac{dz}{dt}_{|t=0} = 6.$$

□

Let $z = f(x, y)$, where both x and y are functions of two variables u, v. That is,

$$z = f(x, y) = f(x(u, v), y(u, v)).$$

Assuming that z is a differentiable function of x and y, and x, y are differentiable functions of u, v, we have

$$\frac{\partial z}{\partial u} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial u}$$

and

$$\frac{\partial z}{\partial v} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial v}.$$

Here, since z is a function of both u and v, that is why we have the partial derivatives (NOT ordinary derivatives) of z with respect to u and v. The above two equations give the chain rule in this case. Figure 2.0.1 helps us to memorize the chain rule. In this figure, u and v are replaced by s and t respectively.

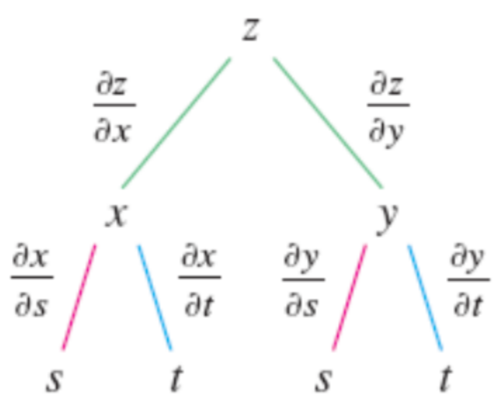

Figure 2.0.1: Chain rule

The general case: Let $z = f(x_1, x_2, \ldots, x_n)$ and for each $i \in \{1, \ldots, n\}$, let $x_i = x_i(t_1, \ldots, t_m)$. Then

$$\frac{\partial z}{\partial t_j} = \frac{\partial z}{\partial x_1}\frac{\partial x_1}{\partial t_j} + \cdots + \frac{\partial z}{\partial x_n}\frac{\partial x_n}{\partial t_j},$$

where $j \in \{1, \ldots, m\}$.
In other words,

$$\frac{\partial z}{\partial t_j} = \sum_{i=1}^{n} \frac{\partial z}{\partial x_i}\frac{\partial x_i}{\partial t_j}.$$

Example 2.0.2. Let $z = f(x, y)$, where $x = r^2 + s^2$ and $y = 2rs$. Find $\frac{\partial z}{\partial r}$ and $\frac{\partial^2 z}{\partial r^2}$.
We know

$$\frac{\partial z}{\partial r} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial r}.$$

This implies that

$$\frac{\partial z}{\partial r} = 2r\frac{\partial f}{\partial x} + 2s\frac{\partial f}{\partial y}.$$

Again,

$$\frac{\partial^2 z}{\partial r^2} = \frac{\partial}{\partial r}\left(\frac{\partial z}{\partial r}\right)$$

$$= \frac{\partial}{\partial r}\left(2r\frac{\partial f}{\partial x} + 2s\frac{\partial f}{\partial y}\right)$$

By the product formula for derivatives, the above expression becomes

$$= 2\frac{\partial f}{\partial x} + 2r\frac{\partial}{\partial r}\left(\frac{\partial f}{\partial x}\right) + 0 + 2s\frac{\partial}{\partial r}\left(\frac{\partial f}{\partial y}\right)$$

By chain rule, this becomes

$$= 2\frac{\partial f}{\partial x} + 2r\left[\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)\frac{\partial x}{\partial r} + \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)\frac{\partial y}{\partial r}\right] + 2s\left[\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)\frac{\partial x}{\partial r} + \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right)\frac{\partial y}{\partial r}\right]$$

$$= 2\frac{\partial f}{\partial x} + 2r\left[2r\frac{\partial^2 f}{\partial x^2} + 2s\frac{\partial^2 f}{\partial y \partial x}\right] + 2s\left[2r\frac{\partial^2 f}{\partial x \partial y} + 2s\frac{\partial^2 f}{\partial y^2}\right]$$

Assuming that the mixed derivatives are continuous, we have from Clairaut's theorem that the above expression equals

$$= 2\frac{\partial f}{\partial x} + 4r^2\frac{\partial^2 f}{\partial x^2} + 4s^2\frac{\partial^2 f}{\partial y^2} + 8rs\frac{\partial^2 f}{\partial x \partial y}.$$

□

3 Directional derivatives

Let $z = f(x, y)$ be a function of two variables x and y. Let (x_0, y_0) be a point in the domain of definition of f. Let $z_0 = f(x_0, y_0)$.

Recall that in case of partial derivatives, we simply drew 2 planes (namely, the planes $x = x_0$ and $y = y_0$) passing through the point $P' = (x_0, y_0, 0)$. And these 2 planes had cut the surface (corresponding to $z = f(x, y)$) into 2 curves, and then we drew tangent lines along those two curves at the point $P = (x_0, y_0, z_0)$. The slopes of these 2 tangent lines became the partial derivatives of f with respect to x and y respectively.

Now suppose that instead of the fixed x and y directions, we have to find the rate of change of $z = f(x, y)$ along an arbitrary direction. For this, we have to consider a plane which is vertical to the xy-plane, along that particular direction. This plane will also intersect the surface (corresponding to $z = f(x, y)$) in a curve, and this curve will pass through the point P. Let Q' be a point in the neighbourhood of P' along that direction. Corresponding to the point Q', let Q denote the point on the curve. We draw a tangent at the point $P = (x_0, y_0, z_0)$ to the curve. The slope of this tangent will give us the rate of change of $z = f(x, y)$ along that direction. This rate of change is nothing but the **directional derivative** of $z = f(x, y)$ in the direction $\hat{u}$, where $\hat{u}$ is a unit vector in the direction in which we were seeking the rate of change. Figure 3.0.1 explains the above phenomenon.

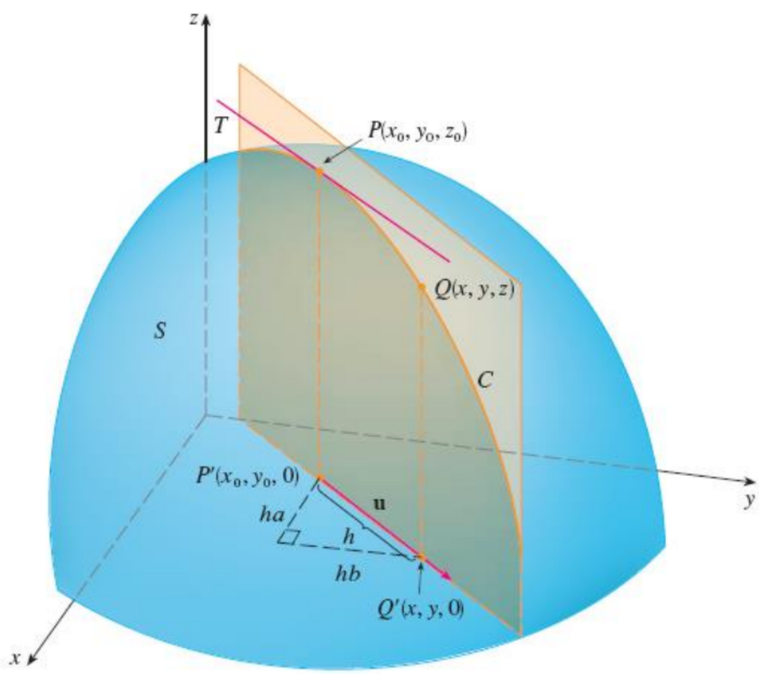

Figure 3.0.1: The directional derivative

Let $|\overrightarrow{P'Q'}| = h$ and $\hat{u} =< a, b >$. If the unit vector $\hat{u}$ makes an angle θ with the x-axis, then we also have $\hat{u} =< \cos\theta, \sin\theta >$. So, it follows that $a = \cos\theta$ and $b = \sin\theta$. Now $\hat{u}$ is a unit vector in the direction of $\overrightarrow{P'Q'}$. So if $\hat{u}$ makes an angle θ with the x-axis, then

$$\overrightarrow{P'Q'} = |\overrightarrow{P'Q'}|\cos\theta\hat{\imath} + |\overrightarrow{P'Q'}|\sin\theta\hat{\jmath}$$

$$= h\cos\theta\hat{\imath} + h\sin\theta\hat{\jmath} = ha\hat{\imath} + hb\hat{\jmath} =< ha, hb >= h\hat{u}.$$

Therefore $x - x_0 = ha$ and $y - y_0 = hb$. So $x = x_0 + ha$, $y = y_0 + hb$, and

$$\frac{\Delta z}{h} = \frac{z - z_0}{h} = \frac{f(x_0 + ha, y_0 + hb) - f(x_0, y_0)}{h}.$$

If we take the limit as $h \to 0$, we obtain the rate of change of z in the direction of u, which is called the directional derivative of f in the direction of u.

Definition 3.0.1. The derivative of $z = f(x, y)$ in the direction $\hat{u}$ at the point (x_0, y_0) is denoted by $D_{\hat{u}}f(x_0, y_0)$ and is defined by

$$D_{\hat{u}}f(x_0.y_0) = \lim_{h\to 0}\frac{f(x_0 + ah, y_0 + bh) - f(x_0, y_0)}{h},$$

where $\hat{u} =< a, b >$ (provided the limit exists). This is called the **directional derivative** of f at (x_0, y_0) in the direction $\hat{u}$. $\square$

Theorem 3.0.2. *If f is a differentiable function of x and y, then f has a directional derivative in the direction of any unit vector $\hat{u} =< a, b >$ and*

$$D_{\hat{u}}f(x, y) = f_x(x, y)a + f_y(x, y)b.$$

PROOF: We will prove that

$$D_{\hat{u}}f(x_0, y_0) = f_x(x_0, y_0)a + f_y(x_0, y_0)b.$$

Let us define a function g (of the variable h) as follows:

$$g(h) := f(x_0 + ha, y_0 + hb).$$

Then

$$g'(0) = \lim_{h\to 0}\frac{g(h) - g(0)}{h - 0}$$

$$= \lim_{h\to 0}\frac{f(x_0 + ha, y_0 + hb) - f(x_0, y_0)}{h} = D_{\hat{u}}f(x_0, y_0).$$

Let $x = x_0 + ha$ and $y = y_0 + hb$. Then we have $g(h) = f(x, y)$ and by chain rule,

$$g'(h) = \frac{dg}{dh} = \frac{\partial g}{\partial x}\frac{dx}{dh} + \frac{\partial g}{\partial y}\frac{dy}{dh}.$$

Or

$$g'(h) = f_x(x, y)a + f_y(x, y)b.$$

This implies that

$$g'(0) = f_x(x_0, y_0)a + f_y(x_0, y_0)b$$

as $x = x_0 + ah$ and $y = y_0 + bh$. □

Definition 3.0.3. The **gradient** of f at (x_0, y_0) is denoted by $\nabla f(x_0, y_0)$ and is defined as

$$\nabla f(x_0, y_0) = f_x(x_0, y_0)\hat{\imath} + f_y(x_0, y_0)\hat{\jmath}.$$

□

4 Implicit differentiation

Before discussing this topic called "Implicit differentiation", let us recall the following **chain rule**:

If f is a function of x and y, and x, y in turn are functions of another variable t, then

$$\frac{df}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}.$$

Let us now move on to the topic called "Implicit differentiation".

Suppose F is a function of 2 variables x and y, and y is again a function of the independent variable x. That is,

$$F(x, y) = F(x, y(x)).$$

Now if we have a relation of the type $F(x, y) = 0$ or constant, then (differentiating with respect to x on both the sides) by applying the above chain rule (where the role of the variable t is played by the variable x), we get:

$$\frac{\partial F}{\partial x}\frac{dx}{dx} + \frac{\partial F}{\partial y}\frac{dy}{dx} = 0.$$

This implies that

$$\frac{dy}{dx} = -\frac{\partial F/\partial x}{\partial F/\partial y},$$

provided $\frac{\partial F}{\partial y} \neq 0$.
In this way, we can find out $\frac{dy}{dx}$ in terms of $\frac{\partial F}{\partial x}$ and $\frac{\partial F}{\partial y}$, although y was not given explicitly in terms of x (to begin with). This is called **implicit differentiation**.

Example 4.0.1. Find $\frac{dy}{dx}$, where $x^3 + y^3 = 3xy$.
Let $F(x,y) = x^3 + y^3 - 3xy$. Then

$$F_x = 3x^2 - 3y \text{ and } F_y = 3y^2 - 3x.$$

So $\frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{x^2-y}{y^2-x}$, provided $y^2 - x \neq 0$. $\square$

5 Directional derivatives continued...

Recall that if f is a function of 2 variables x and y, and $\hat{u} =< a, b >$ is a unit vector. Then the **directional derivative** of f at (x,y) in the direction of $\hat{u}$ is denoted by $D_{\hat{u}}f(x,y)$ and it is defined as

$$D_{\hat{u}}f(x,y) = \lim_{h\to 0} \frac{f(x+ah, y+bh) - f(x,y)}{h}.$$

Recall the follwing theorem as well:

Theorem 5.0.1.

$$D_{\hat{u}}f(x,y) = f_x(x,y)a + f_y(x,y)b =< f_x, f_y > \cdot < a, b > .$$

Recall that the vector $< f_x, f_y >$ was called the **gradient** of f, and is denoted by ∇f. That is,

$$\nabla f = \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j}.$$

Similarly, for a function f of 3 variables x, y, and z, we have

$$\nabla f = \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} + \frac{\partial f}{\partial z}\hat{k}.$$

Example 5.0.2. Let $f(x,y) = x^2y^3 - 4y$. Find $D_{\hat{v}}f$ at $(2,-1)$ in the direction $\vec{v} = 2\hat{i} + 5\hat{j}$.
For this, we firstly need to find the unit vector $\hat{v}$ along $\vec{v}$. Clearly,

$$\hat{v} = \frac{2\hat{i} + 5\hat{j}}{\sqrt{29}}.$$

Also, $\nabla f =< 2xy^3, 3x^2y^2 - 4 >$. So $\nabla f(2,-1) =< -4, 8 >$. Then

$$D_{\hat{v}}f(2,-1) = \nabla f(2,-1) \cdot \hat{v} =< -4, 8 > \cdot < \frac{2}{\sqrt{29}}, \frac{5}{\sqrt{29}} >= \frac{32}{\sqrt{29}}.$$

□

Question: Why is the concept of the gradient vector so significant?
Answer: Because the directional derivative $D_{\hat{u}}f$ is the maximum in the direction of the gradient vector. The explanation is as follows:

$$\begin{aligned} D_{\hat{u}}f(x_0,y_0) &= \nabla f(x_0,y_0) \cdot \hat{u} \\ &= |\nabla f(x_0,y_0)||\hat{u}| \cos\theta, \end{aligned}$$

where θ is the angle between the vectors $\nabla f(x_0,y_0)$ and $\hat{u}$. Since $\hat{u}$ is a unit vector, we therefore get

$$D_{\hat{u}}f(x_0,y_0) = |\nabla f(x_0,y_0)| \cos\theta.$$

From the above equality, it can be easily seen that $D_{\hat{u}}f(x_0,y_0)$ is the maximum when $\theta = 0$. That is, when the directions of $\hat{u}$ coincides with that of $\nabla f(x_0,y_0)$.
Hence we have the following theorem:

Theorem 5.0.3. *Suppose f is a differentiable function of 2 or 3 variables. The maximum value of the directional derivative $D_uf(x)$ is $|\nabla f(x)|$ and it occurs when $\hat{u}$ has the same direction as the gradient vector $\nabla f(x)$.*

Example 5.0.4. Consider a function $T(x,y,z)$ given by

$$T(x,y,z) = \frac{80}{1 + x^2 + 2y^2 + 3z^2}.$$

Here T denotes the temperature (in celcius) and x, y, z denote the coordinates of a place (in metres).

In which direction, T increases fastest at the point $(1, 1, -2)$? Also what is the maximum rate of increase of temperature at the point $(1, 1, -2)$?
It is an easy calculation for you to check that

$$\nabla T(1, 1, -2) =< -\frac{5}{8}, -\frac{10}{8}, \frac{30}{8} > .$$

We know that in the direction of this gradient vector, the directional derivative is maximum. So in the direction of the vector $< -\frac{5}{8}, -\frac{10}{8}, \frac{30}{8} >$, the rate of change of T will be maximum. And the maximum rate of increase equals $|\nabla T|$ celcius/metre (as $|\nabla T| = |\nabla T||\hat{u}| \cos\theta$, $\hat{u}$ is a unit vector, and $\theta = 0$).
□

The following properties of the gradient vector are left to you as homework:

Show that the operation of taking the gradient of a function has the given property. Assume that u and v are differentiable functions of x and y and that a, b are constants.

(a) $\nabla(au + bv) = a\,\nabla u + b\,\nabla v$ (b) $\nabla(uv) = u\,\nabla v + v\,\nabla u$

(c) $\nabla\left(\frac{u}{v}\right) = \frac{v\,\nabla u - u\,\nabla v}{v^2}$ (d) $\nabla u^n = nu^{n-1}\,\nabla u$

The follwoing problem is also a homework for you:

Show that if $z = f(x, y)$ is differentiable at $\mathbf{x}_0 = \langle x_0, y_0 \rangle$, then

$$\lim_{\mathbf{x} \to \mathbf{x}_0} \frac{f(\mathbf{x}) - f(\mathbf{x}_0) - \nabla f(\mathbf{x}_0) \cdot (\mathbf{x} - \mathbf{x}_0)}{|\mathbf{x} - \mathbf{x}_0|} = 0$$

5.1 Method of steepest ascent

Consider a mountain. We can think of the mountain as a surface. This surface can be thought of as a function f of 2 variables x and y, that is, $z = f(x, y)$. Clearly $z = f(x, y)$ is the altitude function. Suppose we are at the base of the mountain (say, at the point $(x_0, y_0, 0)$). If we want to climb the mountain (from the point $(x_0, y_0, 0)$) and reach the peak, then the gradient vector at $(x_0, y_0, 0)$ is going to give us a direction (locally) along which the rate of change of height is the maximum. Suppose we move along that direction. After a while, we will reach another point on the mountain. At the new point, we again consider the gradient vector and move along it. In this way, we are going to get a path. If we follow this path, then we are going to reach the peak of the mountain fastest.

5.2 Directional deriavtive for functions of 3 or more variables

Definition 5.2.1. Let f be a function of 3 variables x, y, z. Then the directional derivative of f at a point (x_0, y_0, z_0) (in its domain) in the direction of a unit vector $\hat{u} =< a, b, c >$ is

$$D_{\hat{u}} f(x_0, y_0, z_0) = \lim_{h \to 0} \frac{f(x_0 + ha, y_0 + hb, z_0 + hc) - f(x_0, y_0, z_0)}{h},$$

if the limit exits. □

If we use vector notation, then we can write the definition of the directional derivative in a more compact form:

$$D_{\mathbf{u}} f(\mathbf{x}_0) = \lim_{h \to 0} \frac{f(\mathbf{x}_0 + h\mathbf{u}) - f(\mathbf{x}_0)}{h},$$

where $\mathbf{x}_0 =< x_0, y_0 >$ if $n = 2$, and $\mathbf{x}_0 =< x_0, y_0, z_0 >$ if $n = 3$. This is reasonable because the vector equation of the line through $\mathbf{x}_0$ in the direction of the vector $\mathbf{u}$ is given by $\mathbf{x} = \mathbf{x}_0 + t\mathbf{u}$ and so $f(\mathbf{x}_0 + h\mathbf{u})$ represents the value of f at a point on this line.

6 Tangent planes to level surfaces

Let us consider a surface as the level surface of a function of 3 variables. Let the level surface be given by the equation $F(x, y, z) = k$. Consider a fixed point $P(x_0, y_0, z_0)$ on this level surface and an arbitrary curve C passing through P. Let

$$\vec{r}(t) =< x(t), y(t), z(t) >, \ t \in [a, b]$$

be a parametric representation of the curve C. Suppose the point $P(x_0, y_0, z_0)$ corresponds to the point $t_0 \in [a, b]$. Then

$$x_0 = x(t_0), \ y_0 = y(t_0), \ z_0 = z(t_0)$$

and $\overrightarrow{OP} = \vec{r}(t_0)$ (where O denotes the origin).
Now for all points (x, y, z) lying on the curve C, the equation

$$F(x(t), y(t), z(t)) = k$$

will be satisfied. Clearly then F becomes a function of t and the ordinary derivative $\frac{dF}{dt} = 0$.
Applying the chain rule, we get:

$$\frac{\partial F}{\partial x}\frac{dx}{dt} + \frac{\partial F}{\partial y}\frac{dy}{dt} + \frac{\partial F}{\partial z}\frac{dz}{dt} = 0.$$

Or in other words,

$$< \frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial F}{\partial z} > \cdot < \frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} >= 0.$$

The above equation is going to be valid for each point on the curve C, and the above equation also says that

$$\nabla F \cdot \vec{r'}(t) = 0.$$

In particular, when $t = t_0$, we have

$$\nabla F(x_0, y_0, z_0) \cdot \vec{r'}(t_0) = 0.$$

That is, the vector $\nabla F(x_0, y_0, z_0)$ is perpendicular to the vector $\vec{r'}(t_0)$.
But $\vec{r'}(t_0)$ is the tangent vector at the point (x_0, y_0, z_0) to the curve C. That means, the gradient vector to the curve at the point P must be perpendicular to the tangent vector. Likewise, if we draw infinitely many curves through the same point P, we will get the complete surface (at least locally). Since the curve C was arbitrary (passing through the point P), it follows that the gradient vector is going to be perpendicular to all the tangent lines to all

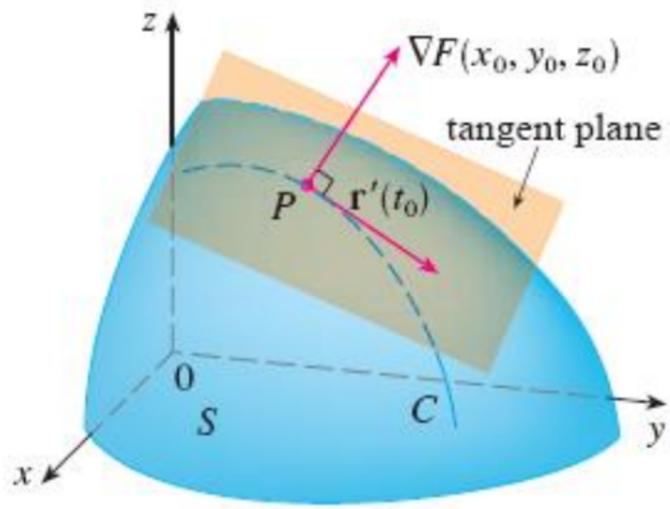

Figure 6.0.1: Tangent planes to level surfaces

these infinitely many curves. These tangent lines lie on the tangent plane to the surface at the point P. This means that the gradient vector at P must be normal to the tangent plane drawn at that point to the surface. This is the significance of the gradient vector. Figure 6.0.1 explains this phenomenon.

We know that the equation of the tangent plane to the surface at the point (x_0, y_0, z_0) is given by

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0,$$

where $\vec{n} =< a, b, c >$ is the normal to tangent plane to the surface at the point (x_0, y_0, z_0). We have also found out just now that the gradient vector is normal to the tangent plane to the surface at (x_0, y_0, z_0). This implies that

$$a = F_x(x_0, y_0, z_0),\ b = F_y(x_0, y_0, z_0),\ c = F_z(x_0, y_0, z_0).$$

So the equation of the tangent plane to the surface $F(x, y, z) = k$ at the point (x_0, y_0, z_0) is given by

$$F_x(x_0, y_0, z_0)(x - x_0) + F_y(x_0, y_0, z_0)(y - y_0) + F_z(x_0, y_0, z_0)(z - z_0) = 0.$$

And the symmetric equation of the normal line is:

$$\frac{x - x_0}{F_x(x_0, y_0, z_0)} = \frac{y - y_0}{F_y(x_0, y_0, z_0)} = \frac{z - z_0}{F_z(x_0, y_0, z_0)}.$$

Example 6.0.1. Find the equation of the tangent plane and the normal line to the surface $\frac{x^2}{4} + y^2 + \frac{z^2}{9} = 3$ at the point $(-2, 1, -3)$. See Figure 6.0.2 for an illustration.

The surface is of the form $F(x, y, z) = k$, where $F(x, y, z) = \frac{x^2}{4} + y^2 + \frac{z^2}{9}$

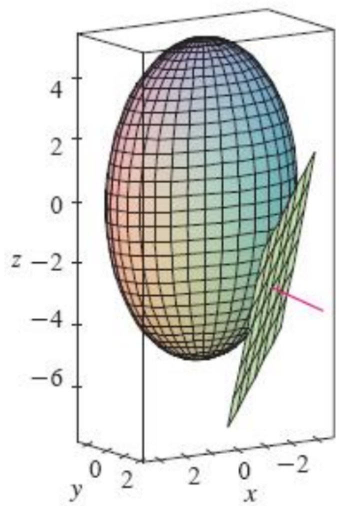

Figure 6.0.2: Figure for Example 6.0.1

and $k = 3$. Therefore,

$$\nabla F = < \frac{x}{2}, 2y, \frac{2z}{9} > .$$

So

$$\nabla F(-2, 1, -3) = < -1, 2, -\frac{2}{3} > .$$

Therefore the equation of the tangent plane at $(-2, 1, -3)$ is given by

$$-1(x + 2) + 2(y - 1) - \frac{2}{3}(z + 3) = 0.$$

And the symmetric equation of the normal line is given by

$$\frac{x + 2}{-1} = \frac{y - 1}{2} = \frac{z + 3}{-2/3}.$$

$\square$

——————-END———————